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CAMBRO-ORDOVICIAN PALEOMAGNETIC POLE POSITION AND RUBIDIUM-STRONTIUM TOTAL ROCK ISOCHRON FOR CHARNOCKITIC ROCKS FROM MIRNYY STATION, EAST ANTARCTICA*

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Charnockitic rocks from Mirnyy Station have yielded a paleomagnetic pole at 1.5°S, 28.5°E with a 95% confidence oval whose semi-axes are 8° and 16°. The rubidium–strontium isotopic date for these rocks is 502 ± 24 my ($\lambda_\beta = 1.47 \times 10^{-11}\,\mathrm{yr}^{-1}$) with an intercept of the total rock isochron line on the $^{87}Sr/^{86}Sr$ axis of 0.7194 ± 0.0007. These results support a reconstruction of Gondwanaland in which the Antarctic Peninsula is closely connected with southern South America.

1. Introduction

Various fits of East Antarctica to Africa have been proposed in Gondwanaland reconstructions. Some [1–3] have advocated a fit which places the Weddell Sea shelf and Princess Martha Coast against Africa, resulting in a small overlap of the Antarctic Peninsula onto Africa. Others [4, 5] have suggested reconstructions in which Antarctica is rotated markedly counterclockwise, leaving the Antarctic Peninsula completely disconnected from the southern Andes, a province to which it is, however, closely related petrologically and structurally [6, 7]. A new paleomagnetic result bearing on this problem is presented in this paper.

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2. Sampling and geology

Samples were collected as drill cores at 21 sites in a 20 km^2 area of outcrop near Mirnyy Station, East Antarctica (66°33′S, 93°01′E). The locations of sample sites are shown in fig. 1. All Mirnyy sites are identified by the prefix RU. Individual specimens for the paleomagnetic study were in the form of cylinders 2.5 cm in diameter and 2.5 cm long, each one designated with a site number, drilling location letter (upper case), core number, and specimen letter (lower case). Drilling locations at each site were separated by several meters and usually more than one core was taken at each location.

The rocks sampled are described by Ravich and others ([8], p. 221) as "charnockites (eulitic and fayalitic granitic rocks)". We did not subject specimens from every site to petrographic examination,

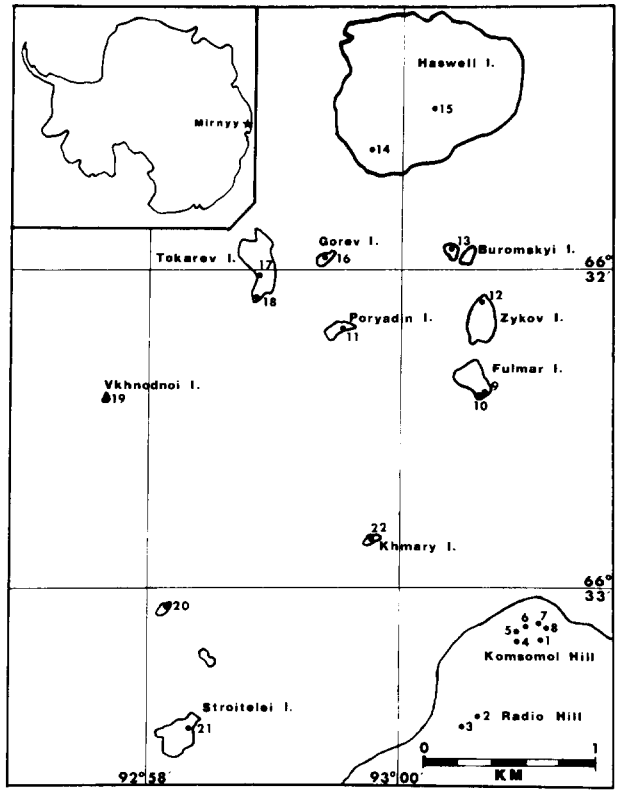


Fig. 1. Map of the vicinity of Mirnyy Station, East Antarctica, showing locations of sample sites.

but ten thin sections that were observed – from sites 1, 3, 8, 11, 12, 14, 15, 17, 18 and 21 – show considerable variation in composition. All contain abundant quartz and feldspar (predominantly plagioclase) and varying amounts of clinopyroxene, orthopyroxene, biotite, apatite and opaques. Hornblende occurs in sections from sites 11, 12, 14, 17 and 18, generally mantling clinopyroxene. Specimens from sites 11, 12, 14, 15, 17, 18 and 21 contain iddingsite, probably as an alteration product of iron-rich olivine. The amount and proportions of the dark minerals vary so widely that no percentages can be given as typical. The range is from about 30% mafics at site 1 to about 3% at site 15. Potassium feldspar, when it occurs, is microcline, usually in perthite. Under the ore microscope the opaques are seen to be magnetite and ilmenite, which often occur together in single grains.

The prevalence of plagioclase over K-feldspar, the ubiquity of biotite and the common predominance of clinopyroxene over orthopyroxene make it questionable whether these rocks can be regarded as true charnockites. They are referred to here as charnockitic rocks.

The geology of the Mirnyy Station area and the complexity of these charnockitic rocks is described in detail by Ravich et al. [8]. Both crystallization from a melt and radical metasomatism seem to have been involved in the final formation of the rocks, which took place during an ultrametamorphic event in early Paleozoic time. The present authors assume that the rubidium – strontium clock was set by the ultrametamorphism and that the stable remanent magnetization dates from that time.

3. Paleomagnetism

Paleomagnetic study has revealed that the natural remanent magnetization (NRM) of specimens from 16 of the 21 sites is unstable when tested by alternating-field (AF) demagnetization. It is common for the NRM directions of unstable specimens to be close

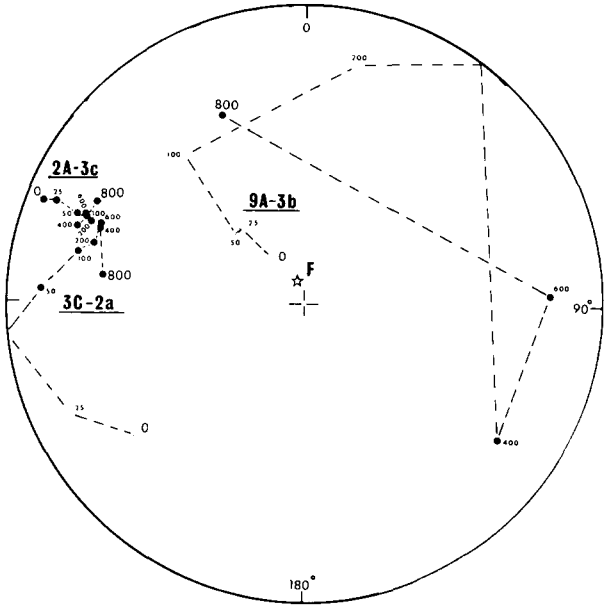


Fig. 2. Change in remanent magnetization vector of two stable specimens (2A–3c and 3C–2a) and one unstable specimen (9A–3b) upon demagnetization at successively higher peak-fields. Open symbols indicate negative inclination (north-seeking direction upward). Star labelled F is the projection of the present-day inclined dipole field vector.

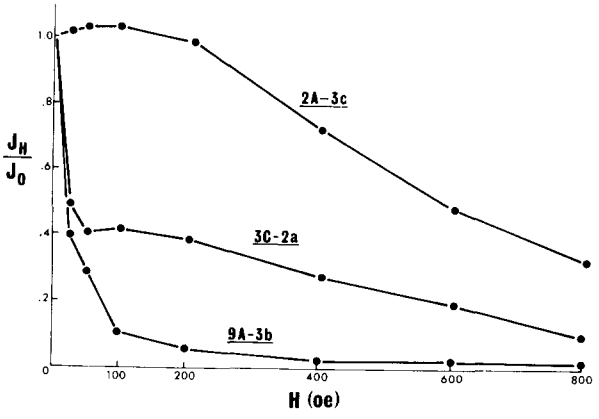


Fig. 3. Decay of remanent magnetization upon alternating field demagnetization for the three specimens depicted in fig. 2. J_H = intensity of RM after cleaning in peak-field H; J_O = intensity of NRM.

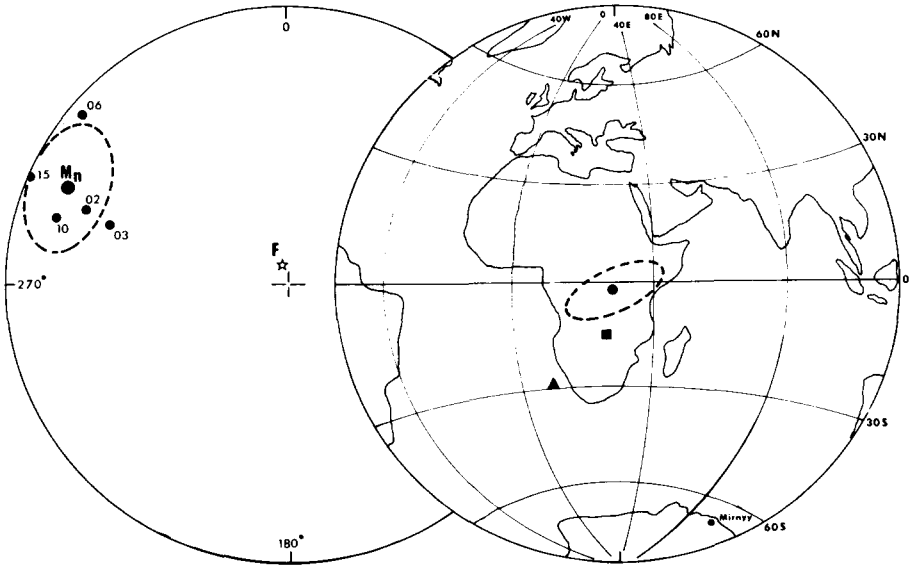


Fig. 4. Equal-area projection (left) of site-mean directions of RM and mean (labelled Mn) of the site-means, with 95% cone of confidence; and plot of virtual geomagnetic poles (right) for Cambro–Ordovician rocks of East Antarctica: dot = this paper; square = result of Nagata and Yama-ai [9]; triangle = result of Zijderveld [10].

Table 1
Paleomagnetic data for five sites in charnockitic rocks, Mirnyy Station, East Antarctica.

Site	AF	N	Declination	Inclination	α_{95}	k	R
RU 2	200	10	291.1	25.7	18.7	7.6	8.82
RU 3	200	9	289.0	34.4	34.4	3.2	6.50
RU 6	100	7	309.7	6.7	37.5	3.5	5.31
RU 10	200	8	287.0	16.5	8.6	42.6	7.84
RU 15	100	3	292.9	0.5	37.3	12.0	2.83
		5	294.2	16.9	15.6	24.8	4.84

AF = peak-field level of magnetic cleaning in Oe; level chosen is that which gives maximum within-site precision.
N = number of specimens.
α_{95} = radius of cone of confidence at 95% level.
k = estimate of precision parameter.
R = magnitude of resultant vector.

to the present-day inclined dipole field direction. The remanent magnetization (RM) directions of specimens from five sites are stable after magnetic cleaning at peak-field levels of 100 or 200 Oe.

The result of AF demagnetization experiments on two stable specimens and one unstable specimen is depicted in figs. 2 and 3, and the paleomagnetic data for the five sites with stable RM are given in table 1. Fig. 4 shows an equal-area plot of the site-mean directions and of the mean vector of the site-means, as well as the position of the virtual geomagnetic pole (VGP) calculated from the mean vector. The VGP is located at 1.5°S, 28.5°E, with an estimated error (95% level of confidence) along the great circle passing through Mirnyy and the VGP of 8.3° and in the orthogonal direction of 16.1°. This position, in south-central Africa, is near that of pole positions for other East Antarctic rocks which have been considered of Cambro-Ordovician age [9, 10].

Remanent magnetism was measured on a PAR SM-1 spinner magnetometer and AF demagnetization was carried out on apparatus designed by Doell and Cox [11]. The usual Fisher [12] statistics were used in the analysis of data.

4. Isotopic age

Five specimens collected from sites within a radius of 650 m were used to determine a rubidium–strontium isochron line. The results of the isotopic analysis are listed in table 2. The age has been calculated from the slope of the isochron line in fig. 5 as 502 ± 24 my

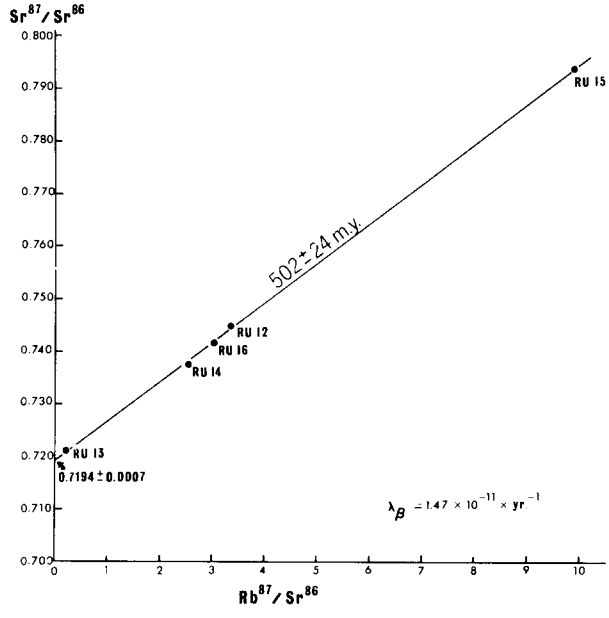


Fig. 5. Total rock isochron diagram of 502 ± 24 my old charnockitic rocks from Mirnyy Station.

by using the decay constant $\lambda_\beta = 1.47 \times 10^{-11} \mathrm{yr}^{-1}$ (532 ± 25 my for $\lambda_\beta = 1.39 \times 10^{-11}\ \mathrm{yr}^{-1}$) and an intercept value of the isochron line on the $^{87}Sr/^{86}$ Sr axis of 0.7194 ± 0.0007. Errors are given at the 95% confidence limits using a least squares technique [13] of fitting data of table 2 to a straight line. Rubidium and strontium isotopic compositions were determined as described in an earlier paper [14]. The Cambro–Ordovician age [15] is assumed to be the final period of isotopic homogenization and closure of the rubidium

Table 2
Rubidium and strontium isotopic analyses of total rock samples of charnockitic material from Mirnyy Station, East Antarctica.

Site	$^{87}Sr/^{86}Sr$[a]	$^{87}Rb/^{86}Sr$	^{87}Rb (μm/g)	^{86}Sr (μm/g)
RU 12	0.7448	3.36	0.638	0.190
RU 13	0.7210	0.201	0.0369	0.184
RU 14	0.7374	2.54	0.531	0.209
RU 15	0.7935	9.91	0.865	0.0873
RU 16	0.7411	3.05	0.569	0.187

[a]Normalized to $^{86}Sr/^{88}Sr$ ratio of 0.1194. At the time of these analyses, the normalized $^{87}Sr/^{86}Sr$ ratio of the M.I.T. standard Eimer and Amend $SrCO_3$ was 0.7081± 0.0003 (mean of four analyses).

and strontium systems of the analyzed rocks. Potassium–argon dates of about 400 to 500 my have been obtained from other charnockitic and gneissic rocks from the Mirnyy area [8].

5. Discussion

In fig. 6 the southern continents and their Cambrian and Ordovician pole positions are shown in two possible early Paleozoic reconstructions. Fig. 6A is after McElhinny and Luck [3] and conforms rather closely to pre-drift relationships proposed by Dietz and Sproll [1] and Smith and Hallam [2]. The Antarctic Peninsula is truncated in this arrangement of the continents in order to avoid an overlap of the peninsula onto Africa. Fig. 6B is a reconstruction which we believe fits the paleomagnetic data from Antarctica more satisfactorily to that from the other continents. This reconstruction is virtually identical with one favored by Dalziel and Elliot [16] who argue for the existence of a continuous cordillera through southern South America and West Antarctica from at least late Paleozoic time.

Paleomagnetic data from Antarctica are still scarce, especially from Paleozoic rocks, so any placement of Antarctica in Gondwanaland on the basis of paleomagnetism must be considered tentative. It is an interesting feature of the reconstruction shown in fig. 6B, however, that the Antarctic Peninsula appears closely linked to southern South America. The history of the Antarctic Peninsula is not well known. If the northern part of the peninsula (Graham Land) is an addition to Antarctica since break-up of Gondwanaland, then there may be little objection to an overlap of the peninsula onto Africa in Gondwanaland reconstructions. Some recent fossil evidence [17], however,

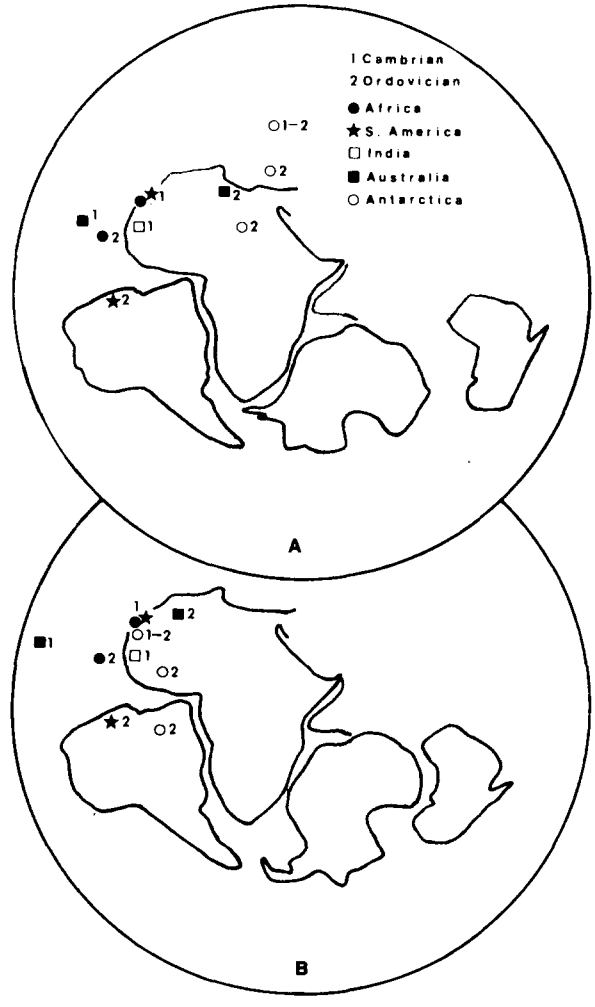


Fig. 6. (A) Reconstruction of Gondwanaland after McElhinny and Luck [3] showing resulting positions of Cambrian and Ordovician paleomagnetic poles. (B) Reconstruction of Gondwanaland which produces a tighter grouping of poles than in A. All pole positions except those for Antarctica are from [3].

suggests that Precambrian rocks occur in Graham Land. A final reconstruction of Gondwanaland will be based on detailed geologic comparison, morphological fits of the continental margins, and sea-floor spreading data, as well as on paleomagnetism. Meanwhile, the hypothetical reconstruction presented in fig. 6B deserves consideration.

Acknowledgements

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